

Gas Pressure, the Particle Model and Diffusion — Paper B

MEDIUM • 40 MINUTES • 30 MARKS

NAME

DATE

SCORE / 30

- Answer **every** question in the space provided. Show your working — method marks are awarded even when the final answer is wrong.
- A calculator is allowed, but every number on this paper is designed to work without one.
- Every numerical answer must carry a **unit**. An answer with no unit does not score the final mark.
- Explanations must be written **in terms of particles**. “The particles get bigger”, “the particles expand” and “the particles push harder because they are squashed” score nothing.
- The mark for each question is shown in square brackets on the right.
- Reminder: **pressure = force ÷ area**, measured in pascals (Pa), where $1 \text{ Pa} = 1 \text{ N/m}^2$. Atmospheric pressure at sea level is about 100 000 Pa.

SECTION 1 – THE ATMOSPHERE

- 1.** Explain why the atmosphere exerts a pressure on everything at the Earth’s surface, and state its approximate value at sea level. [3]

- 2.** Atmospheric pressure is 100 000 Pa.

- (a)** Calculate the force the atmosphere exerts on the palm of a hand of area 0.01 m^2 . [2]

- (b)** A desk top measures 1.2 m by 0.6 m. Calculate the force of the atmosphere on it. [1]

(c) Explain why the desk is not crushed. [1]

3. A little water is boiled inside an empty drinks can until steam pours out of the opening. The can is then turned upside down and pushed into cold water. It is crushed almost at once.

(a) State what happens to the steam inside the can when it meets the cold water. [1]

(b) State what happens to the pressure inside the can, and why. [1]

(c) Explain why the can is crushed. Your answer must use the word “atmospheric”. [2]

4. Explain what makes a drink rise up a straw when you drink through it. Your answer must not use the word “suck”. [3]

SECTION 2 – DIFFUSION

5. State what is meant by diffusion, and explain why it happens even when nothing stirs or blows the gas. [3]

- 6.** A bottle of nail varnish remover is opened at one end of a still, closed room. Two minutes later the smell can be detected six metres away.

(a) Name the process that carried the smell across the room. [1]

(b) The particles move at hundreds of metres per second. Explain why the smell took two minutes and not a fraction of a second. [2]

(c) State one change that would make the smell arrive sooner, and give a reason. [1]

- 7.** Explain why diffusion is much faster in a gas than in a liquid, and why it does not happen in a solid. [3]

- 8.** A gas jar of dense orange bromine vapour is placed under a gas jar of colourless air, with a glass lid between them. The lid is removed. State what is seen over the next few minutes, and explain it in terms of particles. [3]

- 9.** A window pane measures 1.5 m by 0.8 m. Atmospheric pressure is 100 000 Pa.

(a) Calculate the area of the pane. [1]

(b) Calculate the force of the atmosphere on one side of it.

[1]

(c) Explain why the pane does not break.

[1]

End of paper. Check every answer has a unit or a form the question actually asked for.